CBSM response to USAID-BHA pilot study report (Weyori et al, 2023) and commentary (Lacey, 2024)
Draft: April 12, 2024

Purpose: CBSM entered into work agreement with IPC to review and provide feedback about a research report concerning three pilot studies fielded in different countries (Myanmar, Ecuador, and the Central African Republic (CAR)) to conduct a modal survey experiment to ascertain differences in outcomes of food insecurity measures between responses collected via face-to-face interviews and responses collected remotely via phone/cell interviews. CBSM was also asked to respond to commentary provided by Michelle Lacey.

Summary: Overall, the three pilot studies and their accompanying analyses showed merit in the face of limited resources to field a survey experiment such as this in a challenging environment with targeted survey participants who may be reticent to cooperate or difficult to contact in the first place. However, there was a general lack of information in this report required to properly assess the strength and validity of the three pilot studies and subsequent data analysis. For instance, there was scant information regarding the construction of the frame used for respondents in each country, and there was little detail provided about differences in survey administration about each treatment aside from the basic modal difference (face-to-face vs. phone) – details about by-mode differences in the survey instruments, interview protocols for field representatives, and cultural considerations taken to inform said protocols would be very helpful context for the reader. Further detailed commentary is provided below in three broad segments: sample/experimental design; study design; and statistical analysis. Responses to Lacey's commentary round out this review document.

Feedback on Pilot Study Report:

Comments regarding sample/experimental design:

- It appears that there is a difference in terminology used by the researchers regarding sample design for the three pilot studies in this report. In all three studies, a sample frame was stated to have been established to conduct each experiment, but all households in the frame were then used in the experiment in some way – thus, no sample was actually drawn and the sample design is, in effect, a census of the frame. There are valid questions raised by the expert reviewers that we would echo regarding the appropriateness of the ad-hoc frame construction relative to a representative sample of the population of interest, but as stated by the study administrators, that was not the aim of these pilot studies.
- If the intent regarding the request to assess the "sample design" is actually in part to be an assessment of the experimental designs for each of the pilot studies, that's within the realm of CBSM to garner an informed opinion. The simple randomized assignment of households into either the Remote (test treatment) or F2F (control treatment) in both the Myanmar and Ecuador pilots seems reasonable enough, though it is curious that there was no attempt to control assignment on key demographic or socioeconomic characteristics to ensure sufficient representation of sub-groups across treatments. The CAR pilot, with its multi-stage treatment assignment method, demonstrates a fine attempt to correct for low-SES participants in the Remote group.
- The other element of sample frame construction worth mentioning is the size of the frame. It is remarkable that the overall response rates from these three pilot studies is so high (60 percent or more). And yet, the size of the resulting response groups in each treatment are small enough such that apparent modal differences in the various food insecurity measures across two of the three pilot studies are not statistically significant (e.g., Table 3). This is most apparent in the

Myanmar pilot where half of the Remote cases were lost, but it can also be observed in the Ecuador pilot. For Ecuador, the effect of doing a second round of sampling to shore up low response rates in the first round brings those results more or less in line with Myanmar – it begs the question of whether an overall increase in the size of the sample frame would allow the experiment to pick up modal deltas as statistically significant rather than noise due to variability.

- The CAR case is intriguing in that the higher disparity of food insecurity levels could partly be attributed to the unique experimental design that provided inexpensive cell phones to Remote-treatment households to enable their participation in the survey. One can reasonably assume that impoverished households that are more likely to experience food insecurity are also less likely to own cell phones with active service, so the administrators' efforts to bring cell phone access to those households have brought into the study data these households' influence in a way that is not reflected in the Myanmar or Ecuador pilots.
- In general, the aforementioned experimental design involving selection of the phone respondents for CAR is described almost as an afterthought in the appendices, but it is in fact a key design decision that should be highlighted and explored with a comparative analysis into differences between the phone and non-phone groups.
- The lack of weighting to address sample skewness and low response rates were noted as a weakness of the study. If the authors could apply known information about the population the frame is based upon to craft rough weights, the sample could be balanced on key demographics (e.g., age) to control for some of the selection bias.

Comments regarding study design:

- One thing the paper doesn't address is any design plans that were made to reduce any potential data collection challenges in either mode, including those that may be cultural. What information were people provided before the remote interview? Did you schedule times for the interviews where you knew the individual would be in locations with good network coverage? Were there any gender considerations made with interviewers either face-to-face or over the phone? Are there other important demographic differences that need to be considered for these countries/regions for effective interviewer-based data collection (age, class, tribes, etc.).
- There needs to be a more thorough discussion of the goals and limitations of the study, particularly regarding the universe being sampled within each country. The footnote on page 2 indicates one limitation within Myanmar. What about Ecuador and CAR? From what universe is the listing being conducted? Are these intended to be national samples? Or samples of smaller populations like WFP beneficiaries? I think it's fine to use special samples given budgets and other limitations, but the reader needs to have an idea of the external validity of the study to a broader population.
- Another important aspect of the design to mention would be anything that was done to reduce or eliminate costs for people who are completing via mobile phones. Explanation of how mobile phone costs work in these different countries would be helpful: Would it cost people who participated via the phone money to participate (e.g. how do mobile phone cost work in these countries - do people pay by the minute? Are the phone users only changed for outgoing calls and therefore completing the survey remotely is at no cost to the respondent?) If it would cost them money to participate, were they compensated for that?

- Several questions about handling of phone numbers from potential respondents should be addressed:
    - How were the phone numbers provided? Were these gathered during listing?
    - What if someone did not give a phone number? Were they automatically assigned to the control group?
    - Or were respondents only eligible for the experiment if they had provided a phone number? If so, that likely skewed the Face-to-face group based on phone ownership/network coverage.

Comments regarding statistical analysis:

- It's not clear whether families of pairwise inference tests in these tabular results have been adjusted for simultaneous multiple comparisons. To use Table 3 as an example, we have in each pilot-study column a 4-way distribution of the food security measure by F2F, by Remote, and the delta between them. To properly assess the modal differences of each category, the researchers must adjust the individual confidence level of each comparison so that the overall confidence level of the family of tests is held firm at 90%, 95%, or whatever the researchers chose to base inference decisions (I don't believe this is ever stated). Also, another way to get at the question of "does mode affect response patterns on food security measures" as presented in this fashion would be a simple goodness-of-fit chi-square test at minimum, with a proper ANOVA exercise or generalized linear model with fixed and random effects as the preferred strategy. Neither seem to have been done in favor of descriptive analysis with basic pairwise inferential comparisons – not bad for initial stages of exploratory work, but more sophisticated methods should follow.
- It was noted that there was limited information about how the households in the study were determined to be "statistically identical" in order to make comparisons between, for instance, the CARI and rCARI indices.
- Regarding Table 1 (page 3): It would be more informative to break the columns of this table into to two components: face-to-face and remote. Then, report the response rate across the two conditions within the three countries and overall. Since there are response rate differences, this table should be followed with something like the tables in Annex I to test the equivalence of demographics across the conditions.
- It's not clear what "balanced households" means. Is this saying that the sample in CAR was only among phone owners regardless of the group? If CAR has one of the lowest mobile phone ownership, then people in CAR who have mobile phones are likely very different from the type of respondents who do not have mobile phones.
- In comparing the food insecurity measures by mode, it is not really clear to the reader what the differences in the questionnaires/protocol were between the phone and face-to-face group. Did you reduce the number of questions? Were the questions modified for mode? What was the average length of the interviews between phone versus face-to-face? It is difficult to interpret the results without having a good understanding of to what degree there were changes in the questionnaire due to mode.
- Why not compare CARI-inc with rCARI-inc in the analysis tables? All other comparisons were made. That might be a cleaner mode comparison if the measure of income source and change is not different across the CARI and rCARI. The FES comparison in Table 4 is somewhat confounded

by the difference in the FES measures. Furthermore, the numbers in these tables should be in percentages to match the text. It's not clear why some numbers are bolded without asterisks.

- More detail would be helpful to help the reader understand the extent of differences. Perhaps the authors are asking about different items in CARI whereas in rCARI they are asking about all expenditures. Just listing the number of items and giving examples would be helpful context to know. Furthermore, it's important to know how the expenditure module was shortened. Were categories combined or were there certain expenditures not asked about in the shortened version?
- Regarding Table 6: did the respondents actually overestimate their consumption, or are the estimates just higher on the phone than in-person? Could there be interviewer effects between asking this in-person versus on the phone? Could there be any social desirability in the responses?
- Regarding Table 7: the directionality of coping strategies across the three pilot studies is interesting (in Myanmar and CAR, phone respondents were more likely to adopt more reducing coping strategies relative to F2F counterparts, but the opposite was true in Ecuador). It is possible there might be social desirability. It would be helpful to know if there are any cultural elements that could explain this difference.
- Regarding Table 8: It is unclear that the apparent differences in emergency coping strategies can be called "overreporting". It is certainly a large difference but there's no evidence cited that respondents are misreporting. Certainly with something like illegal activities, it is highly likely that there is social desirability at play here. It may be easier for respondents to say they are participating in illegal activities to someone on the phone than someone who has shown up to their house for a F2F interview.
- The intent and function of Table 11 (Impact of Survey Modality on Income) is unclear. Is it trying to show agreement between FES and income indicators? If so, we would suggest showing the percent agreement or some other measure like kappa across the countries.
- Due to the loss of half of the expenditure data in the Myanmar sample due to translation errors, we wonder if there were any potential context effects for questions administered after this module that we should be considering in the analysis. Was the FES module not translated at all? Or were there errors within some questions that render the composition of the index unusable?

<u>Response to Lacey Commentary</u>:

- Lacey described three key research questions that the studies ought to address when assessing the effects of the modal experiment upon the different food insecurity measures. While we agree these are important questions to investigate, we contend that the authors of the report did not adequately showcase efforts to address these research questions, instead focusing primarily on basic pairwise comparisons within each study as a largely descriptive analysis.
- We agree with Lacey that the identical construction of FCS, coping, and income measures between phone and face-to-face modes ought to result in easy, straightforward comparison of response profiles among these measures by mode, were it not for the listed confounding factors. These factors – geographic factors, livelihood variation, and enumerator effects – are all important considerations that appear to not have been accounted for in the analysis for the pilot studies.

- Lacey notes that the pilot studies were not intended to yield nationally-representative results, and that the focus was strictly upon the modal effects upon response profiles relative to each of the food insecurity measures. We echo this statement, and we emphasize that the report did not make this important point explicitly. Furthermore, it is concerning that even without a focus on population-level estimation, there was no clear statement of the type of population that a full-scale survey of this type would aim to measure.
- Lacey correctly states that the simple pairwise comparisons in the study are a welcome starting point for analysis but should be followed with more comprehensive analysis such as mixed models with fixed and random effects. We concur with the assertion, as the types of overall statements made in the report cannot be supported with pairwise comparisons alone.
- The summary statement (below) is largely agreeable:

  > *In summary, the experimental designs proposed for data collection are appropriate for investigating the comparability of remote and face-to-face modalities, as well as for determining the impact of replacing the FES with the INC indicator. However, as is typically the case, the devil is in the details. Care must be taken to minimize confounding factors wherever possible, and where unavoidable (such as in the case of enumerators) to randomize as much as possible to enable analysis that can account for such factors. If this is done, then the results of these studies are likely to be both publishable and impactful.*

  However, we are not entirely convinced to Lacey's first sentence since the authors did not supply sufficient evidence to support this claim of appropriateness. It's possible that the work may have been done, but it's not stated explicitly within the report.